```
#include <kipr/wombat.h>
// MOTORS
int lmotor = 3;
int rmotor = 0;
int flicker = 2;

// SPEED
int speed = 100;
int rspeed = 100;
int lspeed = 95;
int fspeed1 = 25;

// COLOR TRACKER
void forwards_until_line();
void flick();
void flick2();
void veerleft();
void veerright();
void forwards();
void forwards_slow();
void forwards_until_cup();
void backwards();
void backwards_slow();
void backwards_timed();
void backwards_until_line();
void veerleft_backwards();
void veerright_backwards();
void turn_right();
void turn_until_line_r();
void turn_left();
void turn_until_line_l();
void rightangle_right_turn();
void rightangle_left_turn();
void turn_fowrads_l();
void drop_off_cup();
void drop_in_cup();

// SENSORS
int front_sensor = 1;
int line_sensor_l = 5; //max 3700
int line_sensor = 0; //max 3200
int on_line = 2200;
int fbutton = 0;

// SERVO POSITIONS
void basket_position_middle();
```

```
void initial_turntable_position();
void initial_turntable_position_2();
void turntable_position_middle();
void shaft_position_up();
void shaft_position_up_slow();
void servo_position_close_drink();
void servo_position_close_cup();
void servo_position_up();
void servo_position_down();
void servo_position_down_track();
void servo_position_over_cup();
void place_in_box();
int open = 1320;
int open_medium = 1100;
int close_drink = 620;
int close_cup = 780;
int height = 500;
int height2 = 1400;
int height3 = 1100;
int height4 = 650;
int basket = 1;
int basket_middle = 1024;
int shaft_up = 1700;
int in_box = 230;
int shaft = 2;
int turntable = 3;
int turntable_middle = 1090;
int servo_pos_turntable = 200;
int servo_pos_shaft = 1700;

int main()
{
   /*basket_position_middle();
   ao();
   initial_turntable_position();
   ao();
   shaft_position_up();
   ao();
   forwards_until_line();
   ao();
   forwards();
   msleep(300);
   ao();
   flick();
   ao();
   forwards_until_line();
```

```
    ao();
    place_in_box();*/
    shaft_position_up_slow();
    ao();
    msleep(300);
    initial_turntable_position_2();
    ao();



    disable_servos();
    return 0;
}

void forwards()
{
        motor(lmotor, lspeed);
        motor(rmotor, rspeed);
}
void forwards_until_line()
{
        while(analog(line_sensor)<= on_line)
    {
      printf("finding line\n");
      forwards();
    }
    ao();
}
void turntable_position_middle()
{
      set_servo_position(turntable, turntable_middle);
    printf("turntable middle!\n");
}
void initial_turntable_position()
{
      while(servo_pos_turntable < turntable_middle)
      {
        set_servo_position(turntable, servo_pos_turntable);
        msleep(20);
        servo_pos_turntable += 10;
      }
}
void initial_turntable_position_2()
{
      while(servo_pos_turntable < turntable_middle)
      {
```

```
            enable_servos();
            printf("working\n");
            set_servo_position(turntable, turntable_middle);
            msleep(50);
            turntable_middle -= 5;

        }
}

void shaft_position_up()
{
        set_servo_position(shaft, shaft_up);
    printf("shaft middle!\n");

}
void shaft_position_up_slow()
{

                        while(in_box < shaft_up)
        {
            enable_servos();
            printf("shaft\n");
            set_servo_position(shaft, shaft_up);
            msleep(100);
            in_box += 10;

        }

}
void basket_position_middle()
{
    set_servo_position(basket, basket_middle);
    printf("basket middle!\n");
}
void flick()
{
            motor(flicker, -fspeed1);
            msleep(1600);
    ao();
    while(digital(fbutton) < 1)
    {
            motor(flicker, fspeed1);
    }
    ao();
}
void flick2()
```

```
{
        motor(flicker, -fspeed1);
        msleep(170);
  ao();
  motor(flicker, fspeed1);
        msleep(170);
  ao();
}
void place_in_box()
{
  while(analog(front_sensor) < 2800)
  {
    forwards();
  }
  ao();
  while(servo_pos_shaft > in_box)
    {
      set_servo_position(shaft, servo_pos_shaft);
      msleep(10);
      servo_pos_shaft = servo_pos_shaft-10;
    }
}
```